

DEBASHIS BERA

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ABOUT

Computer Science | Software Developer | AI & Web Applications | 2× Hackathon Winner | GATE 2026 Qualified

B.Tech CSE graduate (MAKAUT, 2026) specialising in agentic AI systems, LLM-integrated applications, and interpretable ML pipelines. Strong foundation in Data Structures & Algorithms with competitive problem-solving experience in C++. Experienced in building production-ready AI solutions using LangChain, LangGraph, Mem0, Google Gemini AI, and React.js. Proven under pressure - 2× Hackathon Winner with end-to-end project delivery experience. Passionate about solving real-world engineering problems at the intersection of AI and software development.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, JavaScript, HTML5/CSS3, SQL
- **AI/ML & GenAI:** LLMs, RAG, AI Agents, LangChain, LangGraph, Hugging Face, Prompt Engineering, Google Gemini AI
- **Databases & Knowledge Graphs:** Neo4j, Vector Databases(Qdrant DB), Firebase, MySQL
- **Frameworks & Libraries:** React.js, Tailwind CSS, NumPy, Pandas, SciPy, EmailJS, ElevenLabs TTS, Scikit-learn, Pydantic
- **DevOps & Tools:** Docker, Linux/Bash, Git (Advanced), GitHub, System Design, Claude Code CLI, OpenCode CLI
- **Soft Skills:** Adaptability, Problem Solving, Leadership, Communication, Team Collaboration

EDUCATION

MAKAUT, WB, Kalyani *B.Tech, CSE* (GPA: 8.63/10 CGPA) **2022 - July 2026**

- **Coursework:** OS, COA, Computer Networks, DBMS, Automata Theory, DAA, Machine Learning, Explainable AI (X-AI)

Karkai Vivakanda Vidyapith *12th Science*, (GPA: 91.6%)

Panithar G M Siksha Sadan *10th* (GPA: 93.15%)

KEY PROJECTS

Explainable Machine Learning Model for Arrhythmia Detection | [Code](#) | *Academic* **May 2025 - Present**

- Developed a hybrid ECG beat segmentation and classification pipeline, achieving 98% accuracy using interpretable ML models.
- Implemented comprehensive data preprocessing, feature extraction(CWT) techniques, class balancing using **GAN**, classification using **CNN(AlexNet)** and model interpretability using **X-AI(Grad-CAM)** for medical diagnostics of Arrhythmia.
- Based on the **MIT-BIH Arrhythmia DataSet**

Deb Job Agent — Autonomous AI Job Application System | [Code](#) | *Personal* **June 2026**

- Built a **LangGraph multi-agent** pipeline scraping 6 job portals daily via **Playwright** stealth, using **Gemini Flash API** to score job-resume fit and auto-send personalised cold emails with resume via **Gmail SMTP**.
- Engineered a 1,831-contact HR drip campaign (50 AI-personalised emails/day) with MX validation, bounce detection, **mem0** cross-session memory, and a **Flask** real-time monitoring dashboard.

Aayojan.AI – Heritage Event Management Powered by AI | [Demo](#) | *Hackathon* **Sep 2025 - Sep 2025**

- Built an AI-driven cultural event management platform with features like multilingual content generation, smart finance tracking, role-based dashboards, an AI poster maker, and community engagement tools.
- Engineered using **Next.js (App Router)**, **TypeScript**, **Tailwind**, **Prisma**, **MySQL**, **Better Auth**, **Firebase Auth**, **Google Gemini 2.5 AI**, and **Vercel**; delivered a heritage-inspired, responsive, and accessible design system.

Aarav – My Personal AI Assistant | [Code](#) | *Personal* **Feb 2025 - Present**

- Developed an emotionally intelligent AI assistant with real-time voice conversations, task automation, and personality-driven responses.
- Integrated Google Gemini AI, ElevenLabs TTS, and speech recognition with modular communication and cognitive pipelines.

LEADERSHIP & COMMUNITY EXPERIENCE

Hackaut (College Coding Club) | *Language Coordinator* **Nov 2023 - Present**

- Organised technical workshops enhancing communication and coding skills for 100+ students
- Promoted collaborative learning through structured practice sessions

GDSC MAKAUT '23 | *AI & ML Core Member* **Aug 2023 - Jun 2024**

- Facilitated AI/ML sessions focusing on Python libraries, including NumPy and Pandas
- Mentored 100+ students in machine learning concepts and practical implementations

ACHIEVEMENTS

- **Winner — MetaMorph Season-1 (36-hour hackathon): Built Aayojan.AI (winning project)**
- **Hackathon Winner:** Hack4Bengal 3.0: Won "Best Use of Composio" track for Narada Bot
- **GATE 2026 Qualified — CS & IT**
- **Course Certification:**

Decode DSA With C++ - Physics Wallah ([Verify](#)) | Full-stack Generative and Agentic AI with Python ([Verify](#))